

Construction of Kochen–Specker Sets from Mutually Unbiased Bases

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We give a constructive analysis of Kochen–Specker (KS) contextuality at the level of complete orthogonal bases (contexts). In dimension three, three noncomputational mutually unbiased bases directly generate a union of 165 rays and 130 contexts. Exact provenance accounting gives a symmetric decomposition: the three 69-ray lineages share 21 rays and 10 contexts, while each contributes 48 exclusive rays and 40 lineage-specific contexts. Each 69-ray, 50-context lineage is a KS configuration, and this framework clarifies the relation among the Yu–Oh, Harding–Salinas Schmeis, and Cabello constructions. In dimensions four and five, we instead give geometric orthogonality gadgets whose connector constraints force a central ray to be maximally unbiased relative to the computational basis; these forcing statements do not by themselves assert KS uncolorability. For the concrete four-dimensional example, the 20-ray gadget and the Cabello 18-ray set are mutually reconstructible under rank-three orthogonal complementation inside the 24-ray Peres–Mermin eigen-system, although their context hypergraphs have different two-valued-state behavior. The results show why both the ray set and the specified incidence family—and, when completeness is claimed, every context supported by that ray set—are needed in assessments of KS contextuality.

Keywords: Quantum contextuality, Kochen–Specker theorem, Mutually unbiased bases, Quantum logic, Orthogonality hypergraphs, Quantum foundations

I. INTRODUCTION

The Kochen–Specker (KS) theorem shows that certain finite families of quantum propositions do not admit a context-independent assignment of classical truth values compatible with their orthogonality relations [1]. A finite coordinatized KS configuration in the sense used here is therefore specified not only by a set of rays but also by the chosen family of complete bases (contexts) among those rays.

Definition 1 (Rays and contexts) *A ray is a one-dimensional subspace $[v] = \{\lambda v : \lambda \in \mathbb{C} \setminus \{0\}\}$. A context (block) in $\mathbb{C}^D$ is a set of D mutually orthogonal rays spanning the space. In the associated D -uniform hypergraph the rays are vertices and the contexts are hyperedges. A ray is intertwining if it belongs to at least two contexts; a ray occurring in exactly one context is nonintertwining.*

Definition 2 (Two-valued states) *For a context hypergraph (V, E) , a two-valued state is a map $s : V \rightarrow \{0, 1\}$ satisfying $\sum_{v \in C} s(v) = 1$ for every $C \in E$. A coordinatized hypergraph with no two-valued state is a KS configuration. A family of two-valued states is separating if every pair of distinct rays receives different values in at least one state; it is unital if every ray receives the value 1 in at least one state.*

These definitions distinguish the logical KS criterion from resource measures used in inequalities, chromatic

formulations, or nonlocal games [2–4]. Reduced incidence structures can be useful in those settings, but deleting singly occurring rays or omitting available contexts can change the set of two-valued states. For example, the seven-ray subconfiguration discussed in Ref. [5, Fig. 6(a)] has different classical-state behavior from its thirteen-ray completion. We therefore state explicitly which context hypergraph is being analyzed at each step.

The paper has two main contributions:

1. In $\mathbb{C}^3$, three noncomputational mutually unbiased bases (MUBs), together with a fixed family of algebraic generation maps, produce 225 labelled vectors. Projective identification reduces these to 165 rays, and an exhaustive orthogonality search gives 130 contexts. The provenance decomposition is symmetric: 21 rays and 10 contexts are common to all three 69-ray lineages, while each lineage has 48 exclusive rays and 40 lineage-specific contexts. Each lineage contains 69 rays and 50 contexts and is uncolorable.
2. In $D = 4$ and $D = 5$, connector contexts convert orthogonality into linear equations for $|x_j|^2$, forcing a central vector $u = (x_1, \dots, x_D)$ to be maximally unbiased relative to the computational basis. In the concrete $D = 4$ case, the resulting 20-ray set and the Cabello 18-ray KS set are mutually reconstructible inside the 24-ray Peres–Mermin eigensystem [6–10]; nevertheless, their context hypergraphs have different two-valued-state spaces.

The putrit contribution is the uniform exact provenance decomposition and complete context inventory, rather than a claim that the underlying YO or Cabello

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configurations were previously unknown. The higher-dimensional contribution is a forcing mechanism and its incidence analysis, not a standalone KS proof.

Section II relates the qutrit construction to the Yu–Oh, Harding–Salinas Schmeis, and Cabello diagrams and gives the complete ray and context inventories. Section IV separates the higher-dimensional forcing constraints from their coordinate completions and then compares the $D = 4$ gadget with the Cabello set. Appendix A records the qutrit MUB convention used throughout.

II. THE YU–OH–HARDING–SALINAS SCHMEIS–CABELLO GADGET

The 25-ray, 16-context Yu–Oh (YO) configuration supports a state-independent noncontextuality inequality for a qutrit [11–13], but it is not itself a KS configuration under Definition 2 because it has 24 separating two-valued states. Cabello’s “triple YO diagram” [14] combines three YO copies whose center rays form the Fourier basis

$$\mathcal{B}_1 = \{(1, 1, 1), (1, \omega, \omega^2), (1, \omega^2, \omega)\}, \quad \omega = e^{2\pi i/3}.$$

Relative to the computational rim $\mathcal{B}_0 = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$, this is one member of a complete set of four MUBs. Replacing $\mathcal{B}_1$ by either of the other two noncomputational MUBs listed in Appendix A gives an isomorphic 69-ray, 50-context KS configuration. The same incidence pattern appears in the Harding–Salinas Schmeis construction [15, Fig. 2 and Theorem 2.6]; Fig. 1 uses their Greechie-like notation to display the common structure.

a. Guide to the construction. The calculation has four stages. First, the nine seeds are grouped as $\mathcal{B}_1 = \{u_1, u_2, u_3\}$, $\mathcal{B}_2 = \{u_4, u_5, u_6\}$, and $\mathcal{B}_3 = \{u_7, u_8, u_9\}$. Second, the 25 maps in Table I are applied to every seed, producing 225 labelled vectors. Third, scalar multiples are identified projectively, leaving the 165 rays in Table II; all mutually orthogonal triples among those rays are then enumerated, giving the 130 contexts in Table III. Fourth, every ray and context is classified by the complete set of seed lineages from which it can be generated.

This last step is essential. The three 69-ray lineages have a 21-ray intersection, and no ray belongs to exactly two lineages. Each lineage therefore consists of the same 21 common rays plus 48 exclusive rays. At the context level, 10 contexts are common to all three lineages and each lineage contributes 40 additional contexts, so $10 + 3 \cdot 40 = 130$. Thus the provenance structure is symmetric; it is not a 40–4–4 asymmetry. The latter count results if common rays are assigned to the first lineage by label order rather than recorded as common.

We work projectively throughout and use the Harding–Salinas Schmeis labels, augmented where needed by the symbols e_i . The full tables expose the complete input and output inventories, while the following protocol specifies the exact computation and the propositions isolate the conclusions needed later.

b. Exact enumeration protocol. Every coordinate in the qutrit calculation is reduced exactly to $a + b\omega$ with $a, b \in \mathbb{Q}$ and $\omega^2 = -1 - \omega$; no floating-point tolerance is used. A nonzero vector is projectively canonicalized by division by its first nonzero coordinate, after which equality is equality of the two rational coefficients in every coordinate. Distinct canonical representatives are the graph vertices, and an edge joins two vertices precisely when their Hermitian inner product reduces to zero. In $\mathbb{C}^3$, every triangle is a context because three pairwise orthogonal nonzero rays necessarily span the space. Contexts are stored as sorted triples, eliminating duplicate orderings. Lineage intersections and differences are then ordinary set operations on the canonical representatives and context triples. Two-valued states are enumerated by depth-first assignment with immediate rejection when a context has two assigned 1s or, once fully assigned, no 1; only assignments with exactly one 1 in every context are retained. The same exact branch-and-prune test is used below for the finite $D = 4$ context hypergraphs.

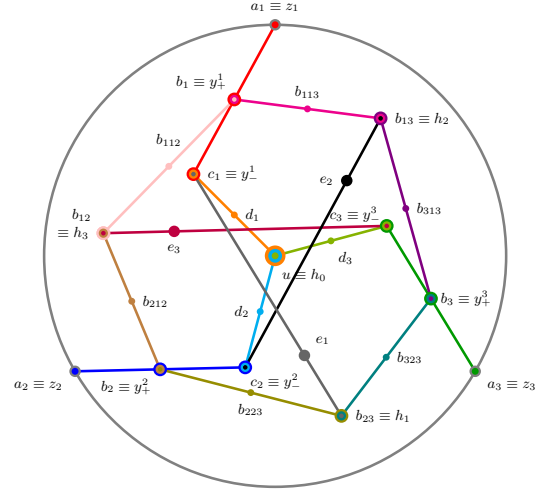


FIG. 1. The Yu–Oh quantum logic [12] drawn in the Harding and Salinas Schmeis 3-uniform hypergraph scheme [15].

III. ENUMERATION OF THE 165 RAYS AND 130 CONTEXTS

The exact projective reduction described in the enumeration protocol above identifies 165 rays among the 225 labelled outputs in Table I. Their canonical representatives are listed in Table II, using $\omega = e^{2\pi i/3}$.

Harding and Salinas Schmeis use a Hermitian cross-product operation to extend orthogonality configurations [15]. To delimit the closure properties of the present finite union, we record that it is not closed under that operation, $u \times v = (u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1)$. For example, $a_{11} \times d_{21} = (1, 0, 0) \times (1, -2, 1) = -(0, 1, 2)$, which is not collinear with any vector of the set of globally non-collinear vectors.

TABLE I. *Systematic generation of the vector set for contextuality analysis.* This table specifies 25 algebraic generation rules by recording their action (rows) on each of the nine initial vectors (columns $u_1 \dots u_9$), for a total of 225 labelled outputs. These initial vectors comprise all three MUBs in $\mathbb{C}^3$: the Fourier basis $\mathcal{B}_1 = \{u_1, u_2, u_3\}$, and two related MUBs, $\mathcal{B}_2 = \{u_4, u_5, u_6\}$ and its complex conjugate $\mathcal{B}_3 = \{u_7, u_8, u_9\}$; each yields a 69-ray, 50-context KS set. The final column counts the number of unique rays produced by each function, highlighting the differing projective variety of the constructions. The 165 unique rays drawn from this set form the labels for identifying orthogonal bases, hyperedges, or contexts. Labelling according to the Harding and Salinas Schmeis scheme [15]; with equivalences to the Yu–Oh numbering scheme [12].

basis	$\mathcal{B}_1$			$\mathcal{B}_2$			$\mathcal{B}_3$			#
label	u_1	u_2	u_3	u_4	u_5	u_6	u_7	u_8	u_9	
$a_1 \equiv z_1$	(1, 0, 0)	(1, 0, 0)	(1, 0, 0)	(1, 0, 0)	(1, 0, 0)	(1, 0, 0)	(1, 0, 0)	(1, 0, 0)	(1, 0, 0)	1
$a_2 \equiv z_2$	(0, 1, 0)	(0, 1, 0)	(0, 1, 0)	(0, 1, 0)	(0, 1, 0)	(0, 1, 0)	(0, 1, 0)	(0, 1, 0)	(0, 1, 0)	1
$a_3 \equiv z_3$	(0, 0, 1)	(0, 0, 1)	(0, 0, 1)	(0, 0, 1)	(0, 0, 1)	(0, 0, 1)	(0, 0, 1)	(0, 0, 1)	(0, 0, 1)	1
$u \equiv h_0$	(1, 1, 1)	(1, ω , ω^2)	(1, ω^2 , ω)	(1, ω , ω)	(1, ω^2 , 1)	(1, 1, ω^2)	(1, ω^2 , ω^2)	(1, ω , 1)	(1, 1, ω)	9
$b_1 \equiv y_+^1$	(0, 1, 1)	(0, ω , ω^2)	(0, ω^2 , ω)	(0, ω , ω)	(0, ω^2 , 1)	(0, 1, ω^2)	(0, ω^2 , ω^2)	(0, ω , 1)	(0, 1, ω)	3
$b_2 \equiv y_+^2$	(1, 0, 1)	(1, 0, ω^2)	(1, 0, ω)	(1, 0, ω)	(1, 0, 1)	(1, 0, ω^2)	(1, 0, ω^2)	(1, 0, 1)	(1, 0, ω)	3
$b_3 \equiv y_+^3$	(1, 1, 0)	(1, ω , 0)	(1, ω^2 , 0)	(1, ω , 0)	(1, ω^2 , 0)	(1, 1, 0)	(1, ω^2 , 0)	(1, ω , 0)	(1, 1, 0)	3
$c_1 \equiv y_-^1$	(0, 1, -1)	(0, ω , $-\omega^2$)	(0, ω^2 , $-\omega$)	(0, ω^2 , $-\omega^2$)	(0, 1, $-\omega$)	(0, ω , -1)	(0, ω , $-\omega$)	(0, 1, $-\omega^2$)	(0, ω^2 , -1)	3
$c_2 \equiv y_-^2$	(-1, 0, 1)	($-\omega$, 0, 1)	($-\omega^2$, 0, 1)	($-\omega^2$, 0, 1)	(-1, 0, 1)	($-\omega$, 0, 1)	($-\omega$, 0, 1)	(-1, 0, 1)	($-\omega^2$, 0, 1)	3
$c_3 \equiv y_-^3$	(1, -1, 0)	(ω^2 , -1, 0)	(ω , -1, 0)	(ω^2 , -1, 0)	(ω , -1, 0)	(1, -1, 0)	(ω , -1, 0)	(ω^2 , -1, 0)	(1, -1, 0)	3
d_1	(-2, 1, 1)	(-2, ω , ω^2)	(-2, ω^2 , ω)	(-2, ω , ω)	(-2, ω^2 , 1)	(-2, 1, ω^2)	(-2, ω^2 , ω^2)	(-2, ω , 1)	(-2, 1, ω)	9
d_2	(1, -2, 1)	(ω^2 , -2, ω)	(ω , -2, ω^2)	(ω^2 , -2, 1)	(ω , -2, ω)	(1, -2, ω^2)	(ω , -2, 1)	(ω^2 , -2, ω^2)	(1, -2, ω)	9
d_3	(1, 1, -2)	(ω , ω^2 , -2)	(ω^2 , ω , -2)	(ω^2 , 1, -2)	(1, ω^2 , -2)	(ω , ω , -2)	(ω , 1, -2)	(1, ω , -2)	(ω^2 , ω^2 , -2)	9
$b_{12} \equiv h_3$	(1, 1, -1)	(1, ω , $-\omega^2$)	(1, ω^2 , $-\omega$)	(ω , ω^2 , $-\omega^2$)	(ω , 1, $-\omega$)	(ω , ω , -1)	(ω^2 , ω , $-\omega$)	(ω^2 , 1, $-\omega^2$)	(ω^2 , ω^2 , -1)	9
$b_{13} \equiv h_2$	(1, -1, 1)	(1, $-\omega$, ω^2)	(1, $-\omega^2$, ω)	(ω , $-\omega^2$, ω^2)	(ω , -1, ω)	(ω , $-\omega$, 1)	(ω^2 , $-\omega$, ω)	(ω^2 , -1, ω^2)	(ω^2 , $-\omega^2$, 1)	9
$b_{23} \equiv h_1$	(-1, 1, 1)	(-1, ω , ω^2)	(-1, ω^2 , ω)	($-\omega$, ω^2 , ω^2)	($-\omega$, 1, ω)	($-\omega$, ω , 1)	($-\omega^2$, ω , ω)	($-\omega^2$, 1, ω^2)	($-\omega^2$, ω^2 , 1)	9
e_1	(1, 2, 1)	(ω^2 , 2, ω)	(ω , 2, ω^2)	(ω , $2\omega^2$, ω^2)	(1, $2\omega^2$, 1)	(ω^2 , $2\omega^2$, ω)	(ω^2 , 2 ω , ω)	(1, 2 ω , 1)	(ω , 2 ω , ω^2)	9
e_2	(1, 1, 2)	(ω , ω^2 , 2)	(ω^2 , ω , 2)	(ω , ω^2 , $2\omega^2$)	(ω^2 , ω , $2\omega^2$)	(1, 1, $2\omega^2$)	(ω^2 , ω , 2 ω)	(ω , ω^2 , 2 ω)	(1, 1, 2 ω)	9
e_3	(2, 1, 1)	(2, ω , ω^2)	(2, ω^2 , ω)	($2\omega^2$, 1, 1)	($2\omega^2$, ω , ω^2)	($2\omega^2$, ω^2 , ω)	(2 ω , 1, 1)	(2 ω , ω^2 , ω)	(2 ω , ω , ω^2)	9
b_{112}	(2, -1, 1)	(2, $-\omega$, ω^2)	(2, $-\omega^2$, ω)	(2, $-\omega$, ω)	(2, $-\omega^2$, 1)	(2, -1, ω^2)	(2, $-\omega^2$, ω^2)	(2, $-\omega$, 1)	(2, -1, ω)	9
b_{212}	(-1, 2, 1)	(-1, 2 ω , ω^2)	(-1, 2 ω^2 , ω)	(-1, 2 ω , ω)	(-1, 2 ω^2 , 1)	(-1, 2, ω^2)	(-1, 2 ω^2 , ω^2)	(-1, 2 ω , 1)	(-1, 2, ω)	9
b_{113}	(2, 1, -1)	(2, ω , $-\omega^2$)	(2, ω^2 , $-\omega$)	(2, ω , $-\omega$)	(2, ω^2 , -1)	(2, 1, $-\omega^2$)	(2, ω^2 , $-\omega^2$)	(2, ω , -1)	(2, 1, $-\omega$)	9
b_{313}	(-1, 1, 2)	(-1, ω , 2 ω^2)	(-1, ω^2 , 2 ω)	(-1, ω , 2 ω)	(-1, ω^2 , 2)	(-1, 1, 2 ω^2)	(-1, ω^2 , 2 ω^2)	(-1, ω , 2)	(-1, 1, 2 ω)	9
b_{223}	(1, 2, -1)	(1, 2 ω , $-\omega^2$)	(1, 2 ω^2 , $-\omega$)	(1, 2 ω , $-\omega$)	(1, 2 ω^2 , -1)	(1, 2, $-\omega^2$)	(1, 2 ω^2 , $-\omega^2$)	(1, 2 ω , -1)	(1, 2, $-\omega$)	9
b_{323}	(1, -1, 2)	(1, $-\omega$, 2 ω^2)	(1, $-\omega^2$, 2 ω)	(1, $-\omega$, 2 ω)	(1, $-\omega^2$, 2)	(1, -1, 2 ω^2)	(1, $-\omega^2$, 2 ω^2)	(1, $-\omega$, 2)	(1, -1, 2 ω)	9

A. Symmetric provenance decomposition

For $r = 1, 2, 3$, let $\mathcal{S}_r$ be the projective ray set generated from the three seeds in $\mathcal{B}_r$, and let $\mathcal{E}_r$ be its set of orthogonal triples.

Proposition 1 (Ray and context inventory) *The generated families satisfy*

$$|\mathcal{S}_r| = 69, \quad |\mathcal{S}_1 \cap \mathcal{S}_2 \cap \mathcal{S}_3| = 21, \quad |\mathcal{S}_r \setminus (\mathcal{S}_s \cup \mathcal{S}_t)| = 48,$$

where $\{r, s, t\} = \{1, 2, 3\}$, and no ray belongs to exactly two lineages. Their union has 165 rays. Likewise, each $\mathcal{E}_r$ contains 50 contexts, 10 contexts are common to all three lineages, and each lineage contributes 40 further contexts. The union therefore contains $10 + 3 \cdot 40 = 130$ contexts. Exactly four contexts in each lineage consist entirely of lineage-exclusive rays.

Proof. This is an exact finite-computation proof using the protocol stated above. Canonicalization gives 69 representatives per lineage and the stated intersections and differences. Exact Hermitian orthogonality gives 50 contexts per lineage and 130 in the union; in dimension three, the graph triangles are contexts because three pairwise orthogonal nonzero rays span $\mathbb{C}^3$. The exact-one constraint solver returns no two-valued state for any of the three 69-ray, 50-context lineages. $\square$

The count of four all-exclusive contexts per lineage is retained because it is the projectively invariant counterpart of the earlier “pure-context” classification. This calculation also explains why a first-occurrence labeling rule can produce a spurious 40–4–4 count: 18 of the 21 common rays were previously displayed with the color of $\mathcal{B}_1$, while the three Cartesian rays were displayed in black. Provenance must instead be assigned from the complete set of generating lineages, which restores the symmetry

TABLE II. The complete set of 165 projectively distinct rays, with six entries per row. The rightmost index records the seed column in Table I. The 21 rays common to all three 69-ray lineages are black. The 48 rays exclusive to $\mathcal{B}_1$, $\mathcal{B}_2$, and $\mathcal{B}_3$ are shown in red, green, and blue, respectively.

$a_{11} = (1, 0, 0)$	$a_{21} = (0, 1, 0)$	$a_{31} = (0, 0, 1)$	$u_1 = (1, 1, 1)$	$u_2 = (1, \omega, \omega^2)$	$u_3 = (1, \omega^2, \omega)$
$u_4 = (1, \omega, \omega)$	$u_5 = (1, \omega^2, 1)$	$u_6 = (1, 1, \omega^2)$	$u_7 = (1, \omega^2, \omega^2)$	$u_8 = (1, \omega, 1)$	$u_9 = (1, 1, \omega)$
$b_{11} = (0, 1, 1)$	$b_{12} = (0, \omega, \omega^2)$	$b_{13} = (0, \omega^2, \omega)$	$b_{21} = (1, 0, 1)$	$b_{22} = (1, 0, \omega^2)$	$b_{23} = (1, 0, \omega)$
$b_{31} = (1, 1, 0)$	$b_{32} = (1, \omega, 0)$	$b_{33} = (1, \omega^2, 0)$	$c_{11} = (0, 1, -1)$	$c_{12} = (0, \omega, -\omega^2)$	$c_{13} = (0, \omega^2, -\omega)$
$c_{21} = (-1, 0, 1)$	$c_{22} = (-\omega, 0, 1)$	$c_{23} = (-\omega^2, 0, 1)$	$c_{31} = (1, -1, 0)$	$c_{32} = (\omega^2, -1, 0)$	$c_{33} = (\omega, -1, 0)$
$d_{11} = (-2, 1, 1)$	$d_{12} = (-2, \omega, \omega^2)$	$d_{13} = (-2, \omega^2, \omega)$	$d_{14} = (-2, \omega, \omega)$	$d_{15} = (-2, \omega^2, 1)$	$d_{16} = (-2, 1, \omega^2)$
$d_{17} = (-2, \omega^2, \omega^2)$	$d_{18} = (-2, \omega, 1)$	$d_{19} = (-2, 1, \omega)$	$d_{21} = (1, -2, 1)$	$d_{22} = (\omega^2, -2, \omega)$	$d_{23} = (\omega, -2, \omega^2)$
$d_{24} = (\omega^2, -2, 1)$	$d_{25} = (\omega, -2, \omega)$	$d_{26} = (1, -2, \omega^2)$	$d_{27} = (\omega, -2, 1)$	$d_{28} = (\omega^2, -2, \omega^2)$	$d_{29} = (1, -2, \omega)$
$d_{31} = (1, 1, -2)$	$d_{32} = (\omega, \omega^2, -2)$	$d_{33} = (\omega^2, \omega, -2)$	$d_{34} = (\omega^2, 1, -2)$	$d_{35} = (1, \omega^2, -2)$	$d_{36} = (\omega, \omega, -2)$
$d_{37} = (\omega, 1, -2)$	$d_{38} = (1, \omega, -2)$	$d_{39} = (\omega^2, \omega^2, -2)$	$b_{121} = (1, 1, -1)$	$b_{122} = (1, \omega, -\omega^2)$	$b_{123} = (1, \omega^2, -\omega)$
$b_{124} = (\omega, \omega^2, -\omega^2)$	$b_{125} = (\omega, 1, -\omega)$	$b_{126} = (\omega, \omega, -1)$	$b_{127} = (\omega^2, \omega, -\omega)$	$b_{128} = (\omega^2, 1, -\omega^2)$	$b_{129} = (\omega^2, \omega^2, -1)$
$b_{131} = (1, -1, 1)$	$b_{132} = (1, -\omega, \omega^2)$	$b_{133} = (1, -\omega^2, \omega)$	$b_{134} = (\omega, -\omega^2, \omega^2)$	$b_{135} = (\omega, -1, \omega)$	$b_{136} = (\omega, -\omega, 1)$
$b_{137} = (\omega^2, -\omega, \omega)$	$b_{138} = (\omega^2, -1, \omega^2)$	$b_{139} = (\omega^2, -\omega^2, 1)$	$b_{231} = (-1, 1, 1)$	$b_{232} = (-1, \omega, \omega^2)$	$b_{233} = (-1, \omega^2, \omega)$
$b_{234} = (-\omega, \omega^2, \omega^2)$	$b_{235} = (-\omega, 1, \omega)$	$b_{236} = (-\omega, \omega, 1)$	$b_{237} = (-\omega^2, \omega, \omega)$	$b_{238} = (-\omega^2, 1, \omega^2)$	$b_{239} = (-\omega^2, \omega^2, 1)$
$e_{11} = (1, 2, 1)$	$e_{12} = (\omega^2, 2, \omega)$	$e_{13} = (\omega, 2, \omega^2)$	$e_{14} = (\omega, 2\omega^2, \omega^2)$	$e_{15} = (1, 2\omega^2, 1)$	$e_{16} = (\omega^2, 2\omega^2, \omega)$
$e_{17} = (\omega^2, 2\omega, \omega)$	$e_{18} = (1, 2\omega, 1)$	$e_{19} = (\omega, 2\omega, \omega^2)$	$e_{21} = (1, 1, 2)$	$e_{22} = (\omega, \omega^2, 2)$	$e_{23} = (\omega^2, \omega, 2)$
$e_{24} = (\omega, \omega^2, 2\omega^2)$	$e_{25} = (\omega^2, \omega, 2\omega^2)$	$e_{26} = (1, 1, 2\omega^2)$	$e_{27} = (\omega^2, \omega, 2\omega)$	$e_{28} = (\omega, \omega^2, 2\omega)$	$e_{29} = (1, 1, 2\omega)$
$e_{31} = (2, 1, 1)$	$e_{32} = (2, \omega, \omega^2)$	$e_{33} = (2, \omega^2, \omega)$	$e_{34} = (2\omega^2, 1, 1)$	$e_{35} = (2\omega^2, \omega, \omega^2)$	$e_{36} = (2\omega^2, \omega^2, \omega)$
$e_{37} = (2\omega, 1, 1)$	$e_{38} = (2\omega, \omega^2, \omega)$	$e_{39} = (2\omega, \omega, \omega^2)$	$b_{1121} = (2, -1, 1)$	$b_{1122} = (2, -\omega, \omega^2)$	$b_{1123} = (2, -\omega^2, \omega)$
$b_{1124} = (2, -\omega, \omega)$	$b_{1125} = (2, -\omega^2, 1)$	$b_{1126} = (2, -1, \omega^2)$	$b_{1127} = (2, -\omega^2, \omega^2)$	$b_{1128} = (2, -\omega, 1)$	$b_{1129} = (2, -1, \omega)$
$b_{2121} = (-1, 2, 1)$	$b_{2122} = (-1, 2\omega, \omega^2)$	$b_{2123} = (-1, 2\omega^2, \omega)$	$b_{2124} = (-1, 2\omega, \omega)$	$b_{2125} = (-1, 2\omega^2, 1)$	$b_{2126} = (-1, 2, \omega^2)$
$b_{2127} = (-1, 2\omega^2, \omega^2)$	$b_{2128} = (-1, 2\omega, 1)$	$b_{2129} = (-1, 2, \omega)$	$b_{1131} = (2, 1, -1)$	$b_{1132} = (2, \omega, -\omega^2)$	$b_{1133} = (2, \omega^2, -\omega)$
$b_{1134} = (2, \omega, -\omega)$	$b_{1135} = (2, \omega^2, -1)$	$b_{1136} = (2, 1, -\omega^2)$	$b_{1137} = (2, \omega^2, -\omega^2)$	$b_{1138} = (2, \omega, -1)$	$b_{1139} = (2, 1, -\omega)$
$b_{3131} = (-1, 1, 2)$	$b_{3132} = (-1, \omega, 2\omega^2)$	$b_{3133} = (-1, \omega^2, 2\omega)$	$b_{3134} = (-1, \omega, 2\omega)$	$b_{3135} = (-1, \omega^2, 2)$	$b_{3136} = (-1, 1, 2\omega^2)$
$b_{3137} = (-1, \omega^2, 2\omega^2)$	$b_{3138} = (-1, \omega, 2)$	$b_{3139} = (-1, 1, 2\omega)$	$b_{2231} = (1, 2, -1)$	$b_{2232} = (1, 2\omega, -\omega^2)$	$b_{2233} = (1, 2\omega^2, -\omega)$
$b_{2234} = (1, 2\omega, -\omega)$	$b_{2235} = (1, 2\omega^2, -1)$	$b_{2236} = (1, 2, -\omega^2)$	$b_{2237} = (1, 2\omega^2, -\omega^2)$	$b_{2238} = (1, 2\omega, -1)$	$b_{2239} = (1, 2, -\omega)$
$b_{3231} = (1, -1, 2)$	$b_{3232} = (1, -\omega, 2\omega^2)$	$b_{3233} = (1, -\omega^2, 2\omega)$	$b_{3234} = (1, -\omega, 2\omega)$	$b_{3235} = (1, -\omega^2, 2)$	$b_{3236} = (1, -1, 2\omega^2)$
$b_{3237} = (1, -\omega^2, 2\omega^2)$	$b_{3238} = (1, -\omega, 2)$	$b_{3239} = (1, -1, 2\omega)$			

summarized in Proposition 1.

IV. HIGHER-DIMENSIONAL FORCING GADGETS

In $\mathcal{P}(\mathbb{C}^3)$, two independent rays determine a unique orthogonal ray. For $D \geq 4$, their orthogonal complement has dimension $D - 2$, so additional connector contexts are required to obtain a rigid finite construction.

Definition 3 (Unbiased vector) *A normalized vector u is maximally unbiased relative to an orthonormal basis $\mathcal{B}$ if*

$$|\langle e_j, u \rangle|^2 = 1/D \quad (j = 1, \dots, D).$$

For $u = (x_1, \dots, x_D)$ in the computational basis, this is equivalent to $|x_1|^2 = \dots = |x_D|^2$.

The $D = 4$ and $D = 5$ constructions have three logically separate parts: (i) a scaffold of coordinate-supported minors, (ii) connector rays whose orthogonality to the center gives equations in $|x_j|^2$, and (iii) completion vectors that turn the required orthogonal pairs

into full contexts. The propositions below are geometric forcing statements; they do not by themselves assert that the resulting finite incidence structures are KS configurations. Coordinate inventories and exact enumeration checks are given afterward.

A. Notation and conventions

Fix an unknown center vector $u = (x_1, \dots, x_D) \in \mathbb{C}^D$ with $x_j \neq 0$ for all j , and use the standard Hermitian inner product

$$\langle v, w \rangle = \sum_m \overline{v_m} w_m.$$

We use vectors to represent the one-dimensional subspaces they span and freely scale them when forming blocks, with the understanding that they must be normalized to form a literal orthonormal basis.

B. An explicit gadget for $D = 4$

TABLE III. The complete list of 130 orthogonal bases supported by the 165 rays. To avoid conflating shared rays with lineage-exclusive rays, the entries are left uncolored; the invariant provenance counts are stated in Proposition 1.

$\{a_{11}, a_{21}, a_{31}\}$	$\{a_{11}, b_{11}, c_{11}\}$	$\{a_{11}, b_{12}, c_{12}\}$	$\{a_{11}, b_{13}, c_{13}\}$	$\{a_{21}, b_{21}, c_{21}\}$	$\{a_{21}, b_{22}, c_{22}\}$
$\{a_{21}, b_{23}, c_{23}\}$	$\{a_{31}, b_{31}, c_{31}\}$	$\{a_{31}, b_{32}, c_{32}\}$	$\{a_{31}, b_{33}, c_{33}\}$	$\{u_1, u_2, u_3\}$	$\{u_1, c_{11}, d_{11}\}$
$\{u_1, c_{21}, d_{21}\}$	$\{u_1, c_{31}, d_{31}\}$	$\{u_2, c_{12}, d_{12}\}$	$\{u_2, c_{22}, d_{22}\}$	$\{u_2, c_{32}, d_{32}\}$	$\{u_3, c_{13}, d_{13}\}$
$\{u_3, c_{23}, d_{23}\}$	$\{u_3, c_{33}, d_{33}\}$	$\{u_4, u_5, u_6\}$	$\{u_4, c_{11}, d_{14}\}$	$\{u_4, c_{23}, d_{24}\}$	$\{u_4, c_{32}, d_{34}\}$
$\{u_5, c_{12}, d_{15}\}$	$\{u_5, c_{21}, d_{25}\}$	$\{u_5, c_{33}, d_{35}\}$	$\{u_6, c_{13}, d_{16}\}$	$\{u_6, c_{22}, d_{26}\}$	$\{u_6, c_{31}, d_{36}\}$
$\{u_7, u_8, u_9\}$	$\{u_7, c_{11}, d_{17}\}$	$\{u_7, c_{22}, d_{27}\}$	$\{u_7, c_{33}, d_{37}\}$	$\{u_8, c_{13}, d_{18}\}$	$\{u_8, c_{21}, d_{28}\}$
$\{u_8, c_{32}, d_{38}\}$	$\{u_9, c_{12}, d_{19}\}$	$\{u_9, c_{23}, d_{29}\}$	$\{u_9, c_{31}, d_{39}\}$	$\{b_{11}, b_{121}, b_{1121}\}$	$\{b_{11}, b_{124}, b_{1124}\}$
$\{b_{11}, b_{127}, b_{1127}\}$	$\{b_{11}, b_{131}, b_{1131}\}$	$\{b_{11}, b_{134}, b_{1134}\}$	$\{b_{11}, b_{137}, b_{1137}\}$	$\{b_{12}, b_{122}, b_{1122}\}$	$\{b_{12}, b_{125}, b_{1125}\}$
$\{b_{12}, b_{129}, b_{1129}\}$	$\{b_{12}, b_{132}, b_{1132}\}$	$\{b_{12}, b_{135}, b_{1135}\}$	$\{b_{12}, b_{139}, b_{1139}\}$	$\{b_{13}, b_{123}, b_{1123}\}$	$\{b_{13}, b_{126}, b_{1126}\}$
$\{b_{13}, b_{128}, b_{1128}\}$	$\{b_{13}, b_{133}, b_{1133}\}$	$\{b_{13}, b_{136}, b_{1136}\}$	$\{b_{13}, b_{138}, b_{1138}\}$	$\{b_{21}, b_{121}, b_{2121}\}$	$\{b_{21}, b_{125}, b_{2125}\}$
$\{b_{21}, b_{128}, b_{2128}\}$	$\{b_{21}, b_{231}, b_{2231}\}$	$\{b_{21}, b_{235}, b_{2235}\}$	$\{b_{21}, b_{238}, b_{2238}\}$	$\{b_{22}, b_{122}, b_{2122}\}$	$\{b_{22}, b_{126}, b_{2126}\}$
$\{b_{22}, b_{127}, b_{2127}\}$	$\{b_{22}, b_{232}, b_{2232}\}$	$\{b_{22}, b_{236}, b_{2236}\}$	$\{b_{22}, b_{237}, b_{2237}\}$	$\{b_{23}, b_{123}, b_{2123}\}$	$\{b_{23}, b_{124}, b_{2124}\}$
$\{b_{23}, b_{129}, b_{2129}\}$	$\{b_{23}, b_{233}, b_{2233}\}$	$\{b_{23}, b_{234}, b_{2234}\}$	$\{b_{23}, b_{239}, b_{2239}\}$	$\{b_{31}, b_{131}, b_{3131}\}$	$\{b_{31}, b_{136}, b_{3136}\}$
$\{b_{31}, b_{139}, b_{3139}\}$	$\{b_{31}, b_{231}, b_{3231}\}$	$\{b_{31}, b_{236}, b_{3236}\}$	$\{b_{31}, b_{239}, b_{3239}\}$	$\{b_{32}, b_{132}, b_{3132}\}$	$\{b_{32}, b_{134}, b_{3134}\}$
$\{b_{32}, b_{138}, b_{3138}\}$	$\{b_{32}, b_{232}, b_{3232}\}$	$\{b_{32}, b_{234}, b_{3234}\}$	$\{b_{32}, b_{238}, b_{3238}\}$	$\{b_{33}, b_{133}, b_{3133}\}$	$\{b_{33}, b_{135}, b_{3135}\}$
$\{b_{33}, b_{137}, b_{3137}\}$	$\{b_{33}, b_{233}, b_{3233}\}$	$\{b_{33}, b_{235}, b_{3235}\}$	$\{b_{33}, b_{237}, b_{3237}\}$	$\{c_{11}, b_{231}, e_{31}\}$	$\{c_{11}, b_{234}, e_{34}\}$
$\{c_{11}, b_{237}, e_{37}\}$	$\{c_{12}, b_{232}, e_{32}\}$	$\{c_{12}, b_{235}, e_{35}\}$	$\{c_{12}, b_{239}, e_{39}\}$	$\{c_{13}, b_{233}, e_{33}\}$	$\{c_{13}, b_{236}, e_{36}\}$
$\{c_{13}, b_{238}, e_{38}\}$	$\{c_{21}, b_{131}, e_{11}\}$	$\{c_{21}, b_{135}, e_{15}\}$	$\{c_{21}, b_{138}, e_{18}\}$	$\{c_{22}, b_{132}, e_{12}\}$	$\{c_{22}, b_{136}, e_{16}\}$
$\{c_{22}, b_{137}, e_{17}\}$	$\{c_{23}, b_{133}, e_{13}\}$	$\{c_{23}, b_{134}, e_{14}\}$	$\{c_{23}, b_{139}, e_{19}\}$	$\{c_{31}, b_{121}, e_{21}\}$	$\{c_{31}, b_{126}, e_{26}\}$
$\{c_{31}, b_{129}, e_{29}\}$	$\{c_{32}, b_{122}, e_{22}\}$	$\{c_{32}, b_{124}, e_{24}\}$	$\{c_{32}, b_{128}, e_{28}\}$	$\{c_{33}, b_{123}, e_{23}\}$	$\{c_{33}, b_{125}, e_{25}\}$
$\{c_{33}, b_{127}, e_{27}\}$	$\{b_{121}, b_{122}, b_{123}\}$	$\{b_{124}, b_{125}, b_{126}\}$	$\{b_{127}, b_{128}, b_{129}\}$	$\{b_{131}, b_{132}, b_{133}\}$	$\{b_{134}, b_{135}, b_{136}\}$
$\{b_{137}, b_{138}, b_{139}\}$	$\{b_{231}, b_{232}, b_{233}\}$	$\{b_{234}, b_{235}, b_{236}\}$	$\{b_{237}, b_{238}, b_{239}\}$		

First, take the computational (Cartesian) basis

$$\begin{aligned} e_1 &= (1, 0, 0, 0), & e_2 &= (0, 1, 0, 0), \\ e_3 &= (0, 0, 1, 0), & e_4 &= (0, 0, 0, 1), \end{aligned}$$

and place the context $\mathcal{B}_{\text{comp}} = \{e_1, e_2, e_3, e_4\}$ on the rim of Fig. 2(a).

In the second construction stage define, for each unordered pair $\{i, j\} \subset \{1, 2, 3, 4\}$, pair minors from a Levi-Civita contraction:

$$\begin{aligned} v_{12} &= (0, 0, \overline{x_4}, -\overline{x_3}), & v_{13} &= (0, -\overline{x_4}, 0, \overline{x_2}), \\ v_{14} &= (0, \overline{x_3}, -\overline{x_2}, 0), & v_{23} &= (\overline{x_4}, 0, 0, -\overline{x_1}), \\ v_{24} &= (-\overline{x_3}, 0, \overline{x_1}, 0), & v_{34} &= (\overline{x_2}, -\overline{x_1}, 0, 0). \end{aligned}$$

Each v_{ij} vanishes in coordinates i, j and satisfies $\langle v_{ij}, u \rangle = 0$ by direct cancellation.

In the third step, choose a vector w_{ij} with the same support as v_{ij} and orthogonal to it, that is, $\langle v_{ij}, w_{ij} \rangle = 0$:

$$\begin{aligned} w_{12} &= (0, 0, x_3, x_4), & w_{13} &= (0, x_2, 0, x_4), \\ w_{14} &= (0, x_2, x_3, 0), & w_{23} &= (x_1, 0, 0, x_4), \\ w_{24} &= (x_1, 0, x_3, 0), & w_{34} &= (x_1, x_2, 0, 0). \end{aligned}$$

For each of the $\binom{4}{2} = 6$ pairs $\{i, j\}$, the set

$$\mathcal{B}_{ij} = \{e_i, e_j, v_{ij}, w_{ij}\}$$

is an orthogonal basis (block, context). Indeed, the assumption $x_m \neq 0$ for every coordinate makes both

complementary-support vectors nonzero; their displayed supports make them orthogonal to e_i, e_j , and their mutual orthogonality makes all four rays independent.

In a fourth step, define three connector vectors

$$\begin{aligned} v_{1234} &= (x_1, x_2, -x_3, -x_4), \\ v_{1324} &= (x_1, -x_2, x_3, -x_4), \\ v_{1423} &= (x_1, -x_2, -x_3, x_4). \end{aligned}$$

Whenever u is orthogonal to one of these connector vectors, the two-dimensional complement of their span supplies completion rays h_1, h_2 . At this stage these are geometric existence witnesses, not rays asserted to belong to a previously fixed finite set; the concrete symmetric realization below chooses them from the displayed 20 rays. Thus the three connector contexts may be written

$$\begin{aligned} \mathcal{C}_{(a)} &= \{u, v_{1234}, h_1^{(a)}, h_2^{(a)}\}, & \mathcal{C}_{(b)} &= \{u, v_{1324}, h_1^{(b)}, h_2^{(b)}\}, \\ \mathcal{C}_{(c)} &= \{u, v_{1423}, h_1^{(c)}, h_2^{(c)}\}. \end{aligned}$$

Proposition 2 ($D = 4$ forcing constraint)

Membership of u and the three connector rays in the contexts above forces u to be maximally unbiased relative to $\mathcal{B}_{\text{comp}}$.

Proof. For example,

$$\langle u, v_{1234} \rangle = |x_1|^2 + |x_2|^2 - |x_3|^2 - |x_4|^2.$$

The other two inner products are obtained by the displayed sign changes, so the three required inner products vanish exactly when

$$|x_1|^2 + |x_2|^2 = |x_3|^2 + |x_4|^2, \quad (1)$$

$$|x_1|^2 + |x_3|^2 = |x_2|^2 + |x_4|^2, \quad (2)$$

$$|x_1|^2 + |x_4|^2 = |x_2|^2 + |x_3|^2. \quad (3)$$

Subtracting pairs of equations gives $|x_2|^2 = |x_3|^2 = |x_4|^2$; substitution gives $|x_1|^2 = |x_2|^2$. After normalization, every squared modulus is $1/4$. $\square$

1. Automatic and equality-induced contexts

We call a context *equality-induced* if its pairwise orthogonality holds only after imposing the equal-modulus relations derived in Proposition 2. Orthogonalities due only to disjoint support, such as $v_{12} \perp v_{34}$, do not constrain u . For the concrete 20-ray representation with $u = (1, 1, 1, 1)$, the computational context, six pair contexts, and three connector contexts give 10 imposed contexts. An exhaustive exact orthogonality search finds exactly seven further contexts. Three follow from disjoint support:

$$\mathcal{B}_{1234} = \{v_{12}, w_{12}, v_{34}, w_{34}\},$$

$$\mathcal{B}_{1324} = \{v_{13}, w_{13}, v_{24}, w_{24}\},$$

$$\mathcal{B}_{1423} = \{v_{14}, w_{14}, v_{23}, w_{23}\}.$$

The remaining four use the equal-modulus relations from Proposition 2:

$$\mathcal{C}_0 = \{u, v_{1234}, v_{1324}, v_{1423}\},$$

$$\mathcal{E}_1 = \{w_{12}, w_{34}, v_{1324}, v_{1423}\},$$

$$\mathcal{E}_2 = \{w_{13}, w_{24}, v_{1234}, v_{1423}\},$$

$$\mathcal{E}_3 = \{w_{14}, w_{23}, v_{1234}, v_{1324}\}.$$

Hence the complete orthogonality hypergraph relative to this chosen 20-ray set has 17 contexts, not 13. Applying the exact-one solver described in the enumeration protocol to all 20 rays and all 17 contexts gives 16 two-valued states, and these states separate every pair of rays; the complete 20-ray hypergraph is therefore not a KS configuration.

2. Concrete $D = 4$ example: $u = (1, 1, 1, 1)$

Take the center vector $u = (1, 1, 1, 1)$ and the rim vectors

$$\begin{aligned} e_1 &= (1, 0, 0, 0), & e_2 &= (0, 1, 0, 0), \\ e_3 &= (0, 0, 1, 0), & e_4 &= (0, 0, 0, 1). \end{aligned}$$

The pair minors and their complementary vectors become:

$$\begin{aligned} v_{12} &= (0, 0, 1, -1), & w_{12} &= (0, 0, 1, 1), \\ v_{13} &= (0, -1, 0, 1), & w_{13} &= (0, 1, 0, 1), \\ v_{14} &= (0, 1, -1, 0), & w_{14} &= (0, 1, 1, 0), \\ v_{23} &= (1, 0, 0, -1), & w_{23} &= (1, 0, 0, 1), \\ v_{24} &= (-1, 0, 1, 0), & w_{24} &= (1, 0, 1, 0), \\ v_{34} &= (1, -1, 0, 0), & w_{34} &= (1, 1, 0, 0). \end{aligned}$$

Each pair block

$$\mathcal{B}_{ij} = \{e_i, e_j, v_{ij}, w_{ij}\}$$

forms an ONB after normalizing v_{ij} and w_{ij} .

The connector vectors are

$$v_{1234} = (1, 1, -1, -1),$$

$$v_{1324} = (1, -1, 1, -1),$$

$$v_{1423} = (1, -1, -1, 1),$$

and direct evaluation gives $\langle u, v_{1234} \rangle = \langle u, v_{1324} \rangle = \langle u, v_{1423} \rangle = 0$.

For this symmetric choice of u (not for a generic coordinate choice), a convenient explicit completion of the connector blocks can be formed using some of the existing minor vectors:

$$C_{(a)} = \{u, v_{1234}, v_{34}, v_{12}\},$$

$$C_{(b)} = \{u, v_{1324}, v_{24}, v_{13}\},$$

$$C_{(c)} = \{u, v_{1423}, v_{23}, v_{14}\}.$$

Each of these sets is composed of four mutually orthogonal vectors. For example, in $C_{(a)}$, the vectors v_{34} and v_{12} have disjoint support, making them orthogonal. The inner products $\langle v_{1234}, v_{34} \rangle$ and $\langle v_{1234}, v_{12} \rangle$ are zero, as are all other inner products involving u . To create literal ONBs, the vectors should be normalized.

Together with $\mathcal{B}_{\text{comp}}$, the six pair contexts and three connector contexts form the 10-context scaffold that enforces Proposition 2. The equal-modulus condition also creates $\mathcal{C}_0$ and the three contexts $\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_3$ listed above. For example, $\langle w_{14}, v_{1234} \rangle = |x_2|^2 - |x_3|^2 = 0$; $\mathcal{E}_3$ is the dotted dark-green curve in Fig. 2(a).

3. Mutual reconstruction inside the Peres–Mermin eigensystem

Both ray sets are subsets of the 24-ray Peres–Mermin eigensystem [9, 16]. They share 15 rays projectively. Table IV gives rank-three orthogonal-complement constructions for the five gadget-only rays and the three Cabello-only rays. Hence each ray set generates the other under the operation “take the unique ray orthogonal to three independent rays.” We therefore call the ray sets *mutually reconstructible under rank-three orthogonal complementation*; this is a statement about ray-set closure, not operational or contextual equivalence.

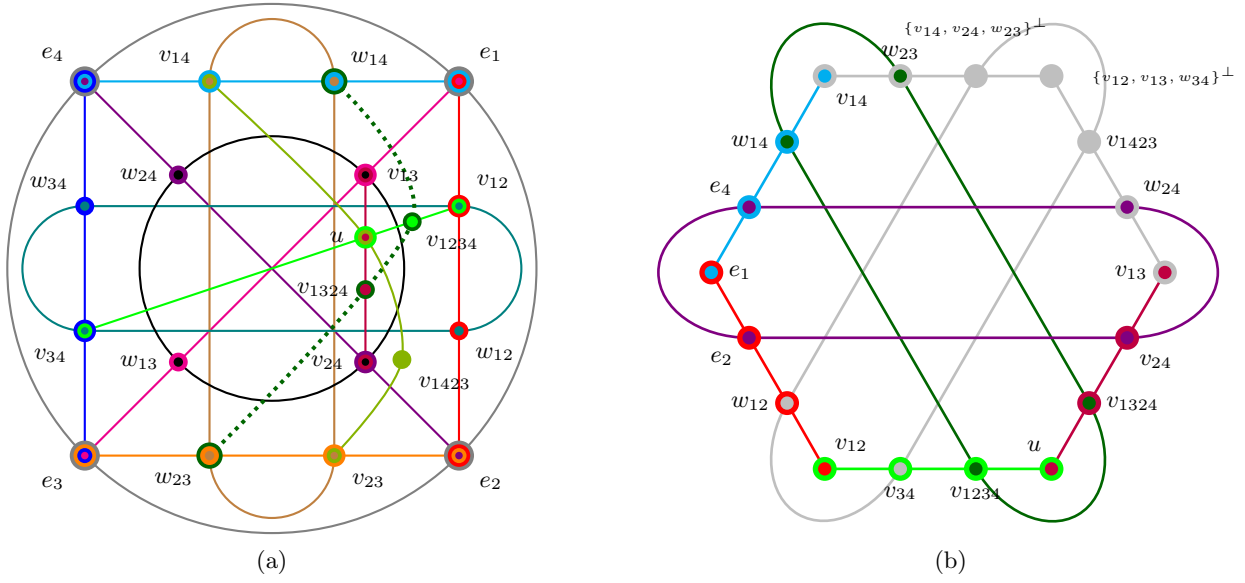


FIG. 2. (a) Four-dimensional extension of the Harding–Salinas Schmeis hypergraph notation [15]. Solid curves show imposed contexts. The dotted dark-green curve is the equality-induced context $\mathcal{E}_3 = \{w_{14}, w_{23}, v_{1234}, v_{1324}\}$; the other contexts of the complete 20-ray orthogonality hypergraph are listed in the text. (b) The Cabello 18-ray, 9-context KS configuration [10], with labels from Ref. [2]. Shared rays are relabeled by the gadget notation; overall nonzero scalar factors are immaterial. The notation $\{v_i, v_j, v_k\}^\perp$ denotes the unique ray orthogonal to the span of three independent rays. The drawings record the stated incidence relations; no lattice-theoretic conclusion is inferred from the planar layout alone.

Mutual reconstructibility does not imply equality of context hypergraphs. The complete 20-ray orthogonality hypergraph relative to its ray set has 17 contexts and 16 separating two-valued states. The Cabello 18-ray hypergraph selects nine contexts and has no two-valued state; its parity argument makes it a KS configuration [10]. The contrast is therefore not a consequence of “more contexts allowing loopholes”—adding constraints cannot create two-valued states on a fixed vertex set—but of the two constructions selecting different rays and different contexts from their common 24-ray parent.

4. Context completion and two-valued states

The context choice within the 20-ray set can itself be tracked explicitly. Let

$$\mathfrak{B}_{10} = \{\mathcal{B}_{\text{comp}}, \mathcal{B}_{ij} \ (1 \leq i < j \leq 4), \mathcal{C}_{(a)}, \mathcal{C}_{(b)}, \mathcal{C}_{(c)}\}$$

be the 10 imposed contexts, and let

$$\mathfrak{B}_{13} = \mathfrak{B}_{10} \cup \{\mathcal{B}_{1234}, \mathcal{B}_{1324}, \mathcal{B}_{1423}\}.$$

The exact-one solver described in the enumeration protocol shows that both hypergraphs have the same 36 two-valued states, and those states are separating. The three disjoint-support contexts therefore record exclusions already obeyed by every state of $\mathfrak{B}_{10}$; they do not remove an assignment. Adding the four equality-induced contexts gives the complete set

$$\mathfrak{B}_{17} = \mathfrak{B}_{13} \cup \{\mathcal{C}_0, \mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_3\},$$

which has 16 separating two-valued states. This nested comparison makes the monotonicity transparent: adding contexts can preserve or reduce the state set, but cannot enlarge it. It also avoids attributing KS contextuality to an incomplete incidence structure merely because some available contexts were omitted.

The apparent asymmetry of Fig. 2(a) is mainly graphical and reflects the selected 20-ray subset. Both it and the Cabello 18-ray set sit inside the more symmetric 24-ray, 24-context Peres configuration [17, 18]; neither subset need inherit all symmetries of the parent.

C. A $D = 5$ scaffold and forcing constraint

Let ε_{abcde} be totally antisymmetric with $\varepsilon_{12345} = 1$. For each triple $\{i, j, k\}$ define

$$(v_{ijk})_m = \sum_{\ell=1}^5 \varepsilon_{mijk\ell} \overline{x_\ell}.$$

The vector v_{ijk} vanishes in coordinates i, j, k and satisfies $\langle v_{ijk}, u \rangle = 0$ by antisymmetry. Because all coordinates of u are nonzero, the complementary two-coordinate restriction is nonzero; one can therefore choose a nonzero w_{ijk} on those coordinates with $v_{ijk} \perp w_{ijk}$. Then

$$\mathcal{B}_{ijk} = \{e_i, e_j, e_k, v_{ijk}, w_{ijk}\}$$

is a scaffold context.

TABLE IV. Comparison of the 20-ray gadget and the Cabello 18-ray set [2, 10]. Both lie in the 24-ray Peres–Mermin eigensystem [6–9, 16]. Fifteen rays are shared projectively. Each of the five gadget-only rays and each of the three Cabello-only rays is the unique orthogonal complement of a three-dimensional span generated by the other set, establishing mutual reconstructibility. Here $\{r_1, r_2, r_3\}^\perp$ denotes the unique ray orthogonal to the span of three independent rays.

Gadget Label	Gadget Vector	Cabello Diagram Label	Vector Construction / Definition
u	$(1, 1, 1, 1)$	v_{56}	$(1, 1, 1, 1)$
e_1	$(1, 0, 0, 0)$	v_{12}	$(1, 0, 0, 0)$
e_2	$(0, 1, 0, 0)$	v_{18}	$(0, 1, 0, 0)$
e_3	$(0, 0, 1, 0)$	(Constructed)	$\{v_{12}, v_{18}, v_{28}\}^\perp$
e_4	$(0, 0, 0, 1)$	v_{28}	$(0, 0, 0, 1)$
v_{12}	$(0, 0, 1, -1)$	v_{16}	$(0, 0, 1, -1)$
v_{13}	$(0, -1, 0, 1)$	v_{45}	$(0, 1, 0, -1)$
v_{14}	$(0, 1, -1, 0)$	v_{23}	$(0, 1, -1, 0)$
v_{23}	$(1, 0, 0, -1)$	(Constructed)	$\{v_{18}, v_{23}, v_{39}\}^\perp$
v_{24}	$(-1, 0, 1, 0)$	v_{58}	$(1, 0, -1, 0)$
v_{34}	$(1, -1, 0, 0)$	v_{67}	$(1, -1, 0, 0)$
w_{12}	$(0, 0, 1, 1)$	v_{17}	$(0, 0, 1, 1)$
w_{13}	$(0, 1, 0, 1)$	(Constructed)	$\{v_{12}, v_{45}, v_{58}\}^\perp$
w_{14}	$(0, 1, 1, 0)$	v_{29}	$(0, 1, 1, 0)$
w_{23}	$(1, 0, 0, 1)$	v_{39}	$(1, 0, 0, 1)$
w_{24}	$(1, 0, 1, 0)$	v_{48}	$(1, 0, 1, 0)$
w_{34}	$(1, 1, 0, 0)$	(Constructed)	$\{v_{28}, v_{16}, v_{67}\}^\perp$
v_{1234}	$(1, 1, -1, -1)$	v_{69}	$(1, 1, -1, -1)$
v_{1324}	$(1, -1, 1, -1)$	v_{59}	$(1, -1, 1, -1)$
v_{1423}	$(1, -1, -1, 1)$	(Constructed)	$\{v_{23}, v_{17}, v_{48}\}^\perp$
(Constructed)	$\{v_{12}, v_{13}, w_{34}\}^\perp$	v_{34}	$(-1, 1, 1, 1)$
(Constructed)	$\{v_{14}, v_{24}, w_{23}\}^\perp$	v_{37}	$(1, 1, 1, -1)$
(Constructed)	$\{v_{13}, v_{23}, w_{12}\}^\perp$	v_{47}	$(1, 1, -1, 1)$

The four connector rays are represented by

$$\begin{aligned} g_1 &= (x_1, 0, 0, 0, -x_5), & g_2 &= (0, x_2, 0, 0, -x_5), \\ g_3 &= (0, 0, x_3, 0, -x_5), & g_4 &= (0, 0, 0, x_4, -x_5). \end{aligned}$$

The four equations suffice because they compare each of $|x_1|^2, \dots, |x_4|^2$ directly with $|x_5|^2$. Whenever $u \perp g_i$, three rays spanning $\text{span}\{u, g_i\}^\perp$ complete the connector context $\mathcal{C}_i = \{u, g_i, h_1^{(i)}, h_2^{(i)}, h_3^{(i)}\}$. As in $D = 4$, these completion rays establish a geometric forcing gadget; no standalone KS claim is made without a separately specified finite ray-context incidence structure.

Proposition 3 ($D = 5$ forcing constraint) *The four connector contexts force a normalized center vector u to be maximally unbiased relative to the computational basis.*

Proof. For $i = 1, 2, 3, 4$,

$$\langle u, g_i \rangle = |x_i|^2 - |x_5|^2.$$

Thus the four orthogonality conditions imply $|x_1|^2 = \dots = |x_5|^2$; normalization makes every squared modulus $1/5$. $\square$

V. SUMMARY AND CONCLUSION

The qutrit construction yields 165 projectively distinct rays and exactly 130 contexts. Correct provenance accounting is symmetric: the three 69-ray, 50-context KS lineages share 21 rays and 10 contexts, and each adds 48 exclusive rays and 40 lineage-specific contexts. This places the YO triple, Harding–Salinas Schmeis, and Cabello descriptions in a common MUB-based framework without assigning shared rays to an arbitrary first lineage.

In $D = 4$ and $D = 5$, the forcing mechanism is explicit at the level of constraints: connector orthogonalities yield linear equations in the squared coordinate moduli and hence maximal unbiasedness. For the concrete $D = 4$ realization, the complete 20-ray orthogonality hypergraph has 17 contexts and 16 separating two-valued states. Its ray set is mutually reconstructible with the Cabello 18-ray set inside the 24-ray Peres–Mermin eigensystem, but the Cabello nine-context hypergraph is uncolorable. Ray-set closure and context incidence are therefore distinct pieces of data.

The main methodological conclusion is that contextuality claims should specify the incidence hypergraph be-

ing tested and, when completeness is intended, enumerate every context supported by the chosen rays. Restricting attention to intertwining rays or to a selected subset of available contexts may define a legitimate reduced scenario, but its two-valued-state properties should not be transferred to a different completion without a separate check.

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Appendix A: Mutually unbiased bases in $\mathbb{C}^3$

For introductions to MUBs, see Refs. [19–24]. Write

$$\omega = e^{2\pi i/3}, \quad \omega^3 = 1, \quad 1 + \omega + \omega^2 = 0, \quad \bar{\omega} = \omega^2.$$

In $D = 3$, a complete MUB family contains four bases [25]. Omitting the common normalization factor

$1/\sqrt{3}$ from the last three bases, we use

$$\mathcal{B}_0 = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}, \quad (\text{A1})$$

$$\mathcal{B}_1 = \{(1, 1, 1), (1, \omega, \omega^2), (1, \omega^2, \omega)\}, \quad (\text{A2})$$

$$\mathcal{B}_2 = \{(1, \omega, \omega), (1, \omega^2, 1), (1, 1, \omega^2)\}, \quad (\text{A3})$$

$$\mathcal{B}_3 = \{(1, \omega^2, \omega^2), (1, \omega, 1), (1, 1, \omega)\}. \quad (\text{A4})$$

Here $\mathcal{B}_0$ is computational and $\mathcal{B}_1$ is the Fourier basis. Direct use of $1 + \omega + \omega^2 = 0$ verifies orthogonality within

each basis and $|\langle b, b' \rangle|^2 = 1/3$ between normalized vectors from different bases. The ordering of $\mathcal{B}_2$ and $\mathcal{B}_3$ agrees with the seed groups in Table I.

For the higher-dimensional argument, no catalogue of a complete MUB family is required: Propositions 2 and 3 use only the criterion $|x_1|^2 = \cdots = |x_D|^2$ for one vector to be maximally unbiased relative to the computational basis.